

LLM-as-a-Judge Report: Repair-understanding MATH500

What was measured?

The judge scores how much the model loses the intended meaning of the question because of corrupted wording. A score of 0 means the meaning is fully understood; 5 means the model is fundamentally lost. This is a semantic-understanding score, not a direct correctness label. The judge returns the score, a verbatim evidence quote when available, and a one-line summary. The tables below use 0–5 scores and include only valid answered traces from `math500_judge_traces.jsonl`.

Examples at different scores

The examples below show the score, final-answer outcome, the judge’s evidence, and its one-line summary. An empty evidence field means that the trace contained no suitable verbatim cue for that dimension.

Trace `typo75_real40`, index 0. **Score:** 0 on the 0–5 scale. **Final-answer outcome:** correct.
Judge evidence:

Alright, so I need to convert the point (0, 3) from rectangular (Cartesian) coordinates to polar coordinates.

One-line summary: The trace led to the scores because it showed a brief hesitation with phrases like ‘Let me confirm that’ and ‘I don’t think I made any mistakes here’, but overall understood the task correctly.

Trace `typo25_real40`, index 4. **Score:** 2 on the 0–5 scale. **Final-answer outcome:** wrong.
Judge evidence:

Okay, so I have this Asymptote graph here, and it’s supposed to show the results of aross-county team’s graining run.

One-line summary: The trace is filled with repeated reconsiderations and second-guessing moments, such as ‘Wait, hold on’ and ‘Maybe each student’s performance is represented by a single point’, indicating significant self-doubt.

Trace typo50_real70, index 184. **Score:** 5 on the 0–5 scale. **Final-answer outcome:** wrong.
Judge evidence:

Okay, so I have this problem where a, b, and c are opposite real numbers. Hmm, opposite real numbers... I think that means they are negatives of each other, right?

One-line summary: The trace is filled with repeated reconsideration and self-doubt due to the ambiguous term 'opposite real numbers' which led to multiple interpretations and confusion.

Aggregate tables

Table 1: Mean score by real-word typo ratio

real_ratio	n	repair_understanding_score_mean
10	1379	0.318
40	1352	0.39
70	1369	0.473

Table 2: Mean score by typo percentage

typo_percentage	n	repair_understanding_score_mean
0	464	0.032
25	1365	0.349
50	1360	0.351
75	1375	0.48

Table 3: Mean score by final-answer correctness

outcome	n	repair_understanding_score_mean
correct	4174	0.268
wrong	390	1.305

Consequences

In this run, more typos are associated with less ability to recover and understand the question meaning: the mean rises from 0.053 for clean traces to 0.580 at 25 percent typos, 0.576 at 50 percent, and 0.764 at 75 percent. Higher scores mean more loss of the intended meaning.